WEATHER AND INCOME'S IMPACT ON CAPITAL BIKESHARE RIDES

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BACKGROUND

IMPORTANCE

- Bikesharing is a popular, low-cost method of transportation in cities
- Transportation is critical
- Easy and cheap transportation methods should be accessible to those who need it the most
- If weather disproportionately affects lower income neighborhoods, transportation eauity could be affected

RESEARCH QUESTION

Does weather have a more extreme effect on Capital Bikeshare ridership in lower income neighborhoods in DC?

DATA

- Capital Bikeshare Data
- National Oceanic and Atmospheric Administration (NOAA)
- Census Income Data

Variable	N	Min	Mean	Max
Rides	148,190	0.00	12.79	190.00
Mean Temp (°F)	148,190	20.00	59.53	89.00
Precipitation (in.)	148,190	0.00	0.10	1.77
Snow (in.)	148,190	0.00	0.04	7.20
Wind Speed	148,190	2.70	8.57	18.30
Income ($10k)	148,190	1.11	12.11	25.00
Poverty %	148,190	0.76	12.53	76.46

392 stations

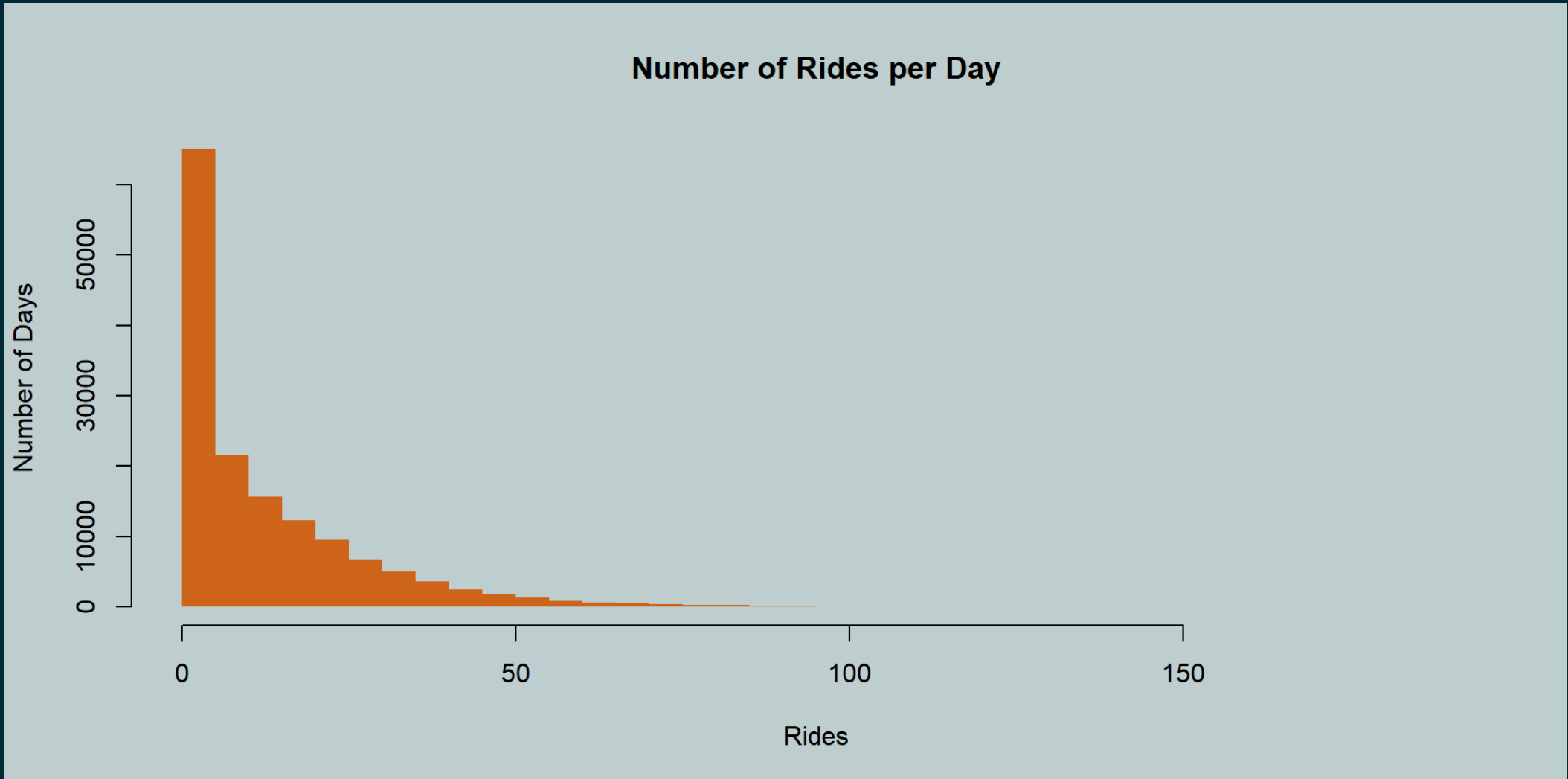
CHALLENGES AND LIMITATIONS WITH THE DATA

- Missing values
  - Removing e-bikes
  - DCA weather station
  - Imputation for wind speed
- Rides outside DC boundaries were removed

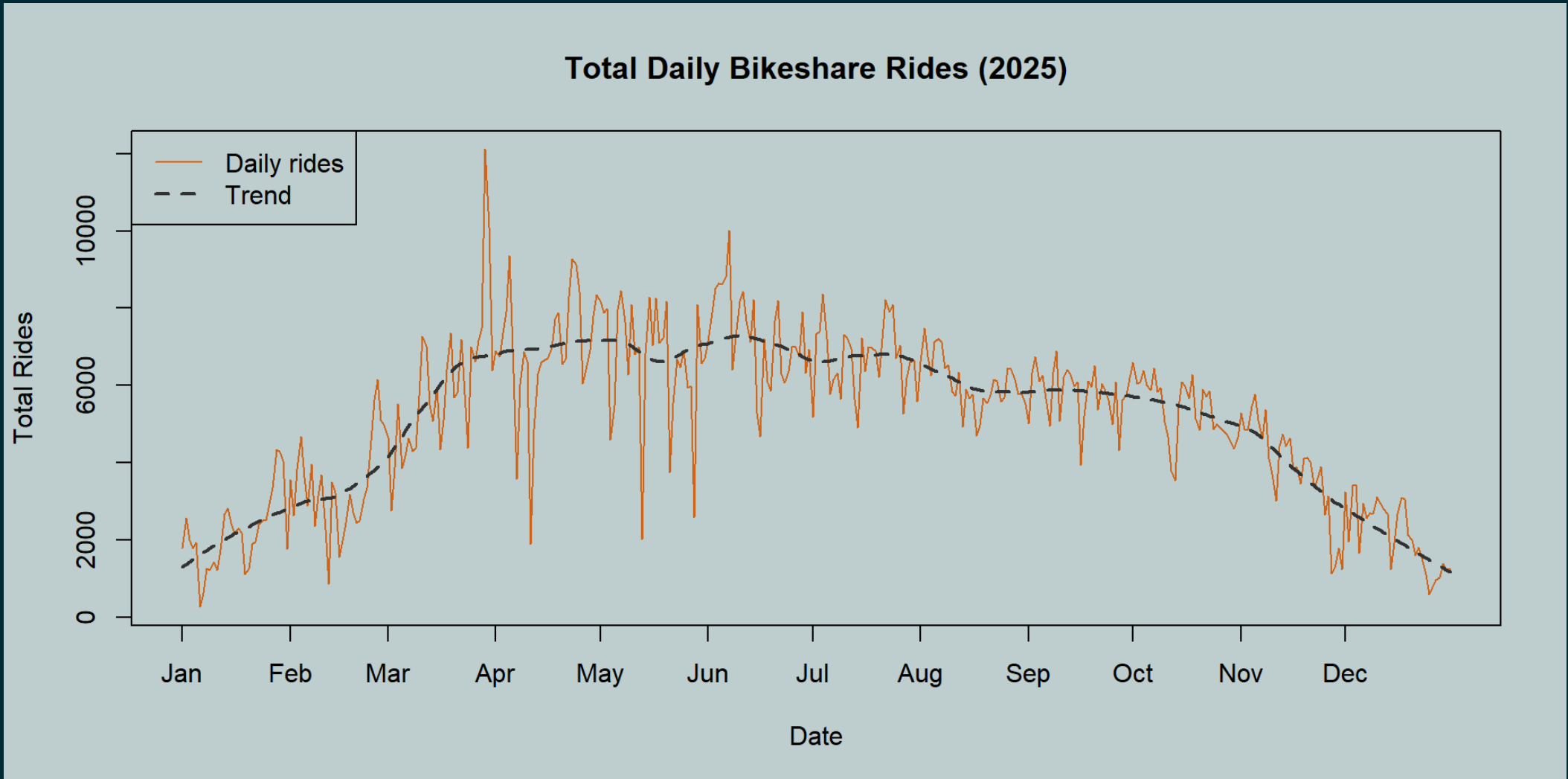
DATA STRUCTURE

Station	Date	Rides	Med. Income	Rain (in.)	Snow (in.)	Mean Temp. (°F)
9th St & Pennsylvania Ave NW	2025-01-01	4	151659	0.00	0.0	46.0
9th St & Pennsylvania Ave NW	2025-01-02	5	151659	0.00	0.0	41.0
9th St & Pennsylvania Ave NW	2025-01-03	2	151659	0.04	0.0	38.5
9th St & Pennsylvania Ave NW	2025-01-04	3	151659	0.00	0.0	30.0
9th St & Pennsylvania Ave NW	2025-01-05	1	151659	0.00	0.0	30.5
9th St & Pennsylvania Ave NW	2025-01-06	0	151659	0.81	7.2	31.0

DAILY RIDES


Number of Rides per Day
Number of Days
0
10000
30000
50000
0
50
100
150
Rides

TIME SERIES


Total Daily Bikeshare Rides (2025)
Daily rides
Trend
Total Rides
0
2000
6000
10000
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Date

MODELING

GOALS

- Want to model a daily ride count using weather, income, and other socioeconomic predictors
  - Poisson model
- Does the effect of weather vary by neighborhood income?
  - Weather x income interactions

POISSON

- Modeling count data
- Predictors relate to ride counts on a log scale
- Coefficients are exponentiated to get Incidence Rate Ratios (IRRs)
- Key assumption: mean = variance

POISSON RESULTS

- Poisson assumption (mean = variance) was badly violated
- Poisson model dispersion statistic = 14.21
  - Standard errors become too small, making results look more significant than they are
- Negative Binomial model adds a parameter to absorb extra variability
- Dispersion drops to 1.17

NEGATIVE BINOMIAL MODEL

- First: full model with all weather × income interactions
  - Only temperature × income was significant
  - Precipitation, snow, and wind interactions were not
- Then fitted a simplified model
  - AIC improved slightly

COMPARISON

- AIC drops dramatically with Negative Binomial
- Dispersion is much better on NB
- RMSE very similar

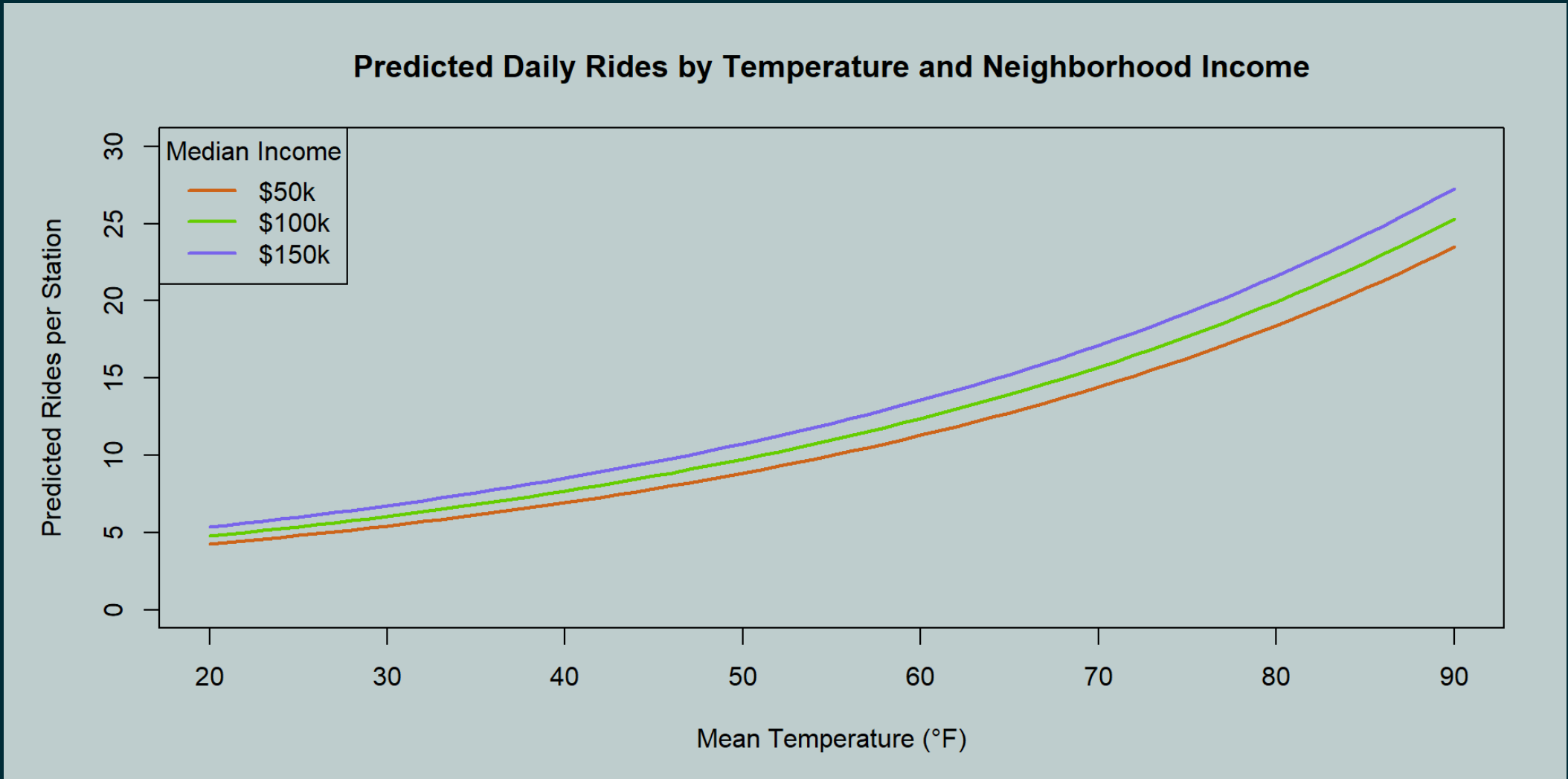
Model	AIC	Dispersion	RMSE
Poisson	2596830	14.21	14.81
Negative Binomial	1025364	1.17	14.94

INTERPRETATION

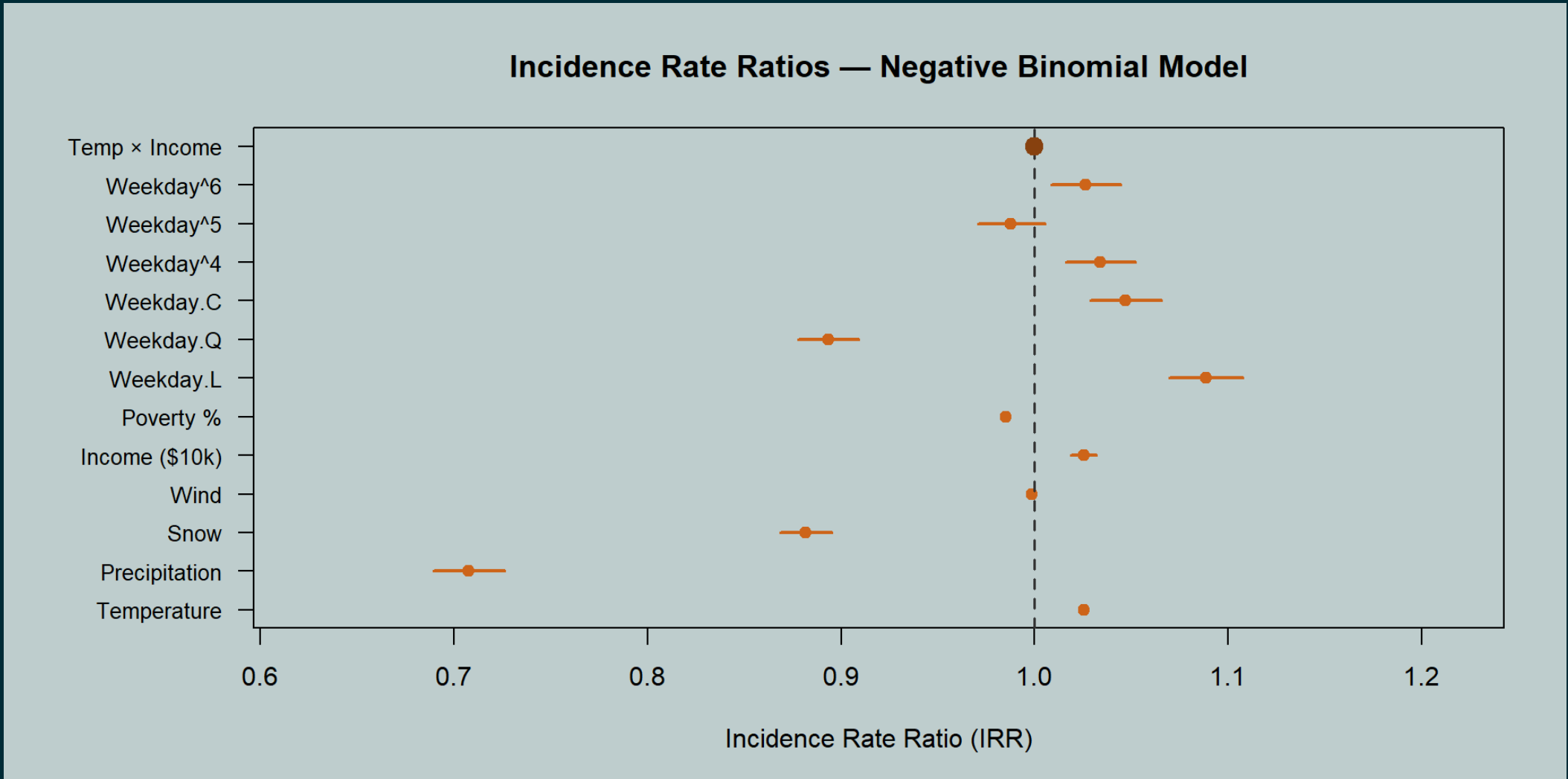
MODEL RESULTS

Term	IRR	SE	CI Low	CI High	p-value	
Intercept	2.686	0.034	2.464	2.929	0.000	***
Temperature	1.025	0.000	1.024	1.027	0.000	***
Precipitation	0.707	0.012	0.689	0.726	0.000	***
Snow	0.882	0.007	0.869	0.895	0.000	***
Wind	0.998	0.001	0.996	1.001	0.120	
Income ($10k)	1.025	0.002	1.019	1.032	0.000	***
Poverty %	0.985	0.000	0.984	0.986	0.000	***
Weekday.L	1.088	0.008	1.070	1.108	0.000	***
Weekday.Q	0.893	0.008	0.878	0.909	0.000	***
Weekday.C	1.047	0.008	1.029	1.065	0.000	***
Weekday^4	1.034	0.008	1.016	1.052	0.000	***
Weekday^5	0.988	0.008	0.971	1.005	0.132	
Weekday^6	1.026	0.008	1.009	1.044	0.001	**
Temp × Income	**1.000**	**0.000**	**1.000**	**1.000**	**0.002**	**

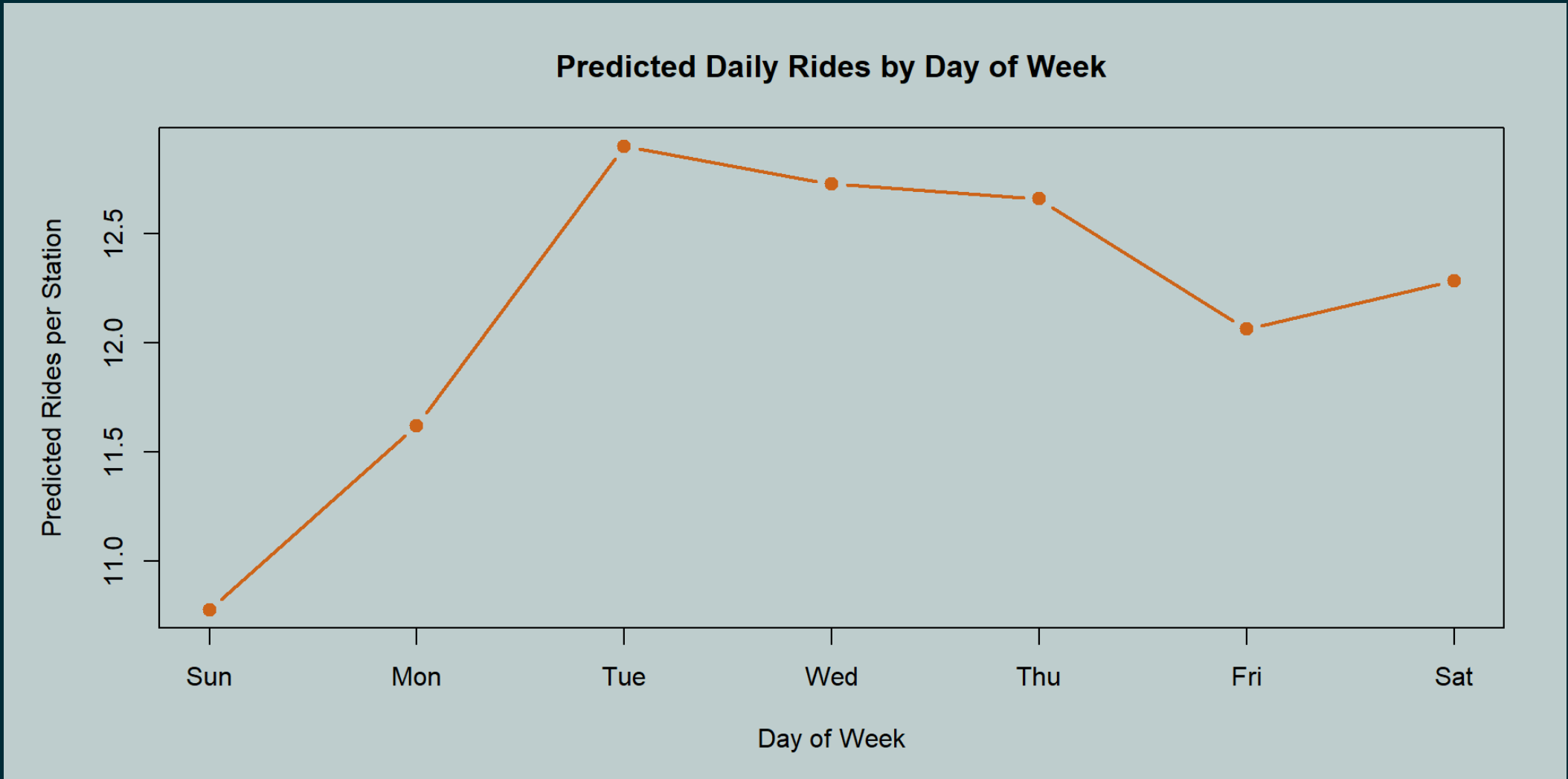
TEMPERATURE X INCOME INTERACTION



COEFFICIENT PLOT



RIDES BY DAY OF WEEK



CONCLUSIONS

- Weather affects ridership broadly
- The temperature x income interaction is minimal but significant
- Precipitation and snow suppress ridership uniformly regardless of income
- Income independently predicts ridership, suggesting access gaps beyond just weather sensitivity

THANK YOU